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High-Yield Formula Sheet, Protocol Matrix & BIT Mesra PYQ Solutions

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⚡ High-Yield DCCN Formulas & Protocol Matrix

Concept	Exact Formula / Rule	Key Insight
Nyquist Capacity	$C = 2B \log_2(M)$ bps	Noiseless channel only.
Shannon Capacity	$C = B \log_2(1 + \text{SNR})$ bps	Convert dB to ratio: $\text{SNR} = 10^{(\text{dB}/10)}$.
Hamming Code Bits	$2^k \geq m + k + 1$	m data bits, k parity bits.
Stop-and-Wait Efficiency	$\eta = \frac{1}{1+2a}$ where $a = \frac{T_{\text{prop}}}{T_{\text{trans}}}$	$T_{\text{trans}} = \frac{L}{B}$, $T_{\text{prop}} = \frac{d}{v}$.
GBN Window Limit	$W_s \leq 2^k - 1, \quad W_r = 1$	Avoids sequence wrap-around ambiguity.
Selective Repeat Limit	$W_s = W_r \leq 2^{k-1}$	Sender and receiver windows are equal.

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